

Leo Chen

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EDUCATION

Santa Clara University

B.S. Electrical and Computer Engineering

Santa Clara, CA

Sept 2022 – Jun 2025

Activities: Institute of Electrical and Electronics Engineers (IEEE), Association of Computing Machinery (ACM)

Coursework: Operating Systems, Computer Architecture, Embedded C & Assembly, Algorithms, Artificial Intelligence

EXPERIENCE

Software Engineer

ProSomnus Sleep Technologies

Jun 2025 – Present

Pleasanton, CA

- Spearheaded full-stack mobile application development for the ProSomnus RPMO2 (FDA Class II oral oximeter), integrating the cross-platform client with cloud backend services and the ProSomnus Oxymetrx ecosystem.
- Architected and deployed high-performance RESTful APIs in Go (Golang) hosted on AWS, designing MySQL schemas to securely ingest, process, and query multi-night patient physiologic telemetry.
- Engineered cross-platform Bluetooth Low Energy (BLE) communication protocols and built custom Python diagnostic tools, ensuring reliable, low-latency telemetry transmission across iOS, Android, Linux, and Windows.
- Implemented secure Over-The-Air (OTA) firmware upgrade workflows and automated internal software tooling, streamlining device manufacturing testing and deployment.
- Authored comprehensive automated test suites across mobile and backend services, ensuring robust code coverage and adhering to FDA medical device software standards (IEC 62304).

Undergraduate Machine Learning Researcher

Santa Clara University

Jun 2024 – May 2025

Santa Clara, CA

- Developed modular vision transformer models using TensorFlow, Keras, and PyTorch on Linux systems.
- Investigated Chain-of-Thought prompting and Backbone Branching to enhance detection transformer capabilities.
- Optimized detection transformers for satellite imagery via architectural layer modifications and hyperparameter tuning.
- Benchmarked models via ablation studies to improve inference throughput and object detection accuracy (mAP).

SKILLS

Languages: Go (Golang), Python, C/C++, SQL (MySQL), Dart, JavaScript/TypeScript, Unix Shell, Verilog

Cloud & Backend: AWS, MySQL, Podman(Docker)

Frameworks & Libraries: Flutter, BLE Protocols (Bleak), PyTorch, TensorFlow, Keras, Angular, NumPy, Matplotlib

Interests: Linux, Vim & Neovim, Open Source, Photography

PROJECTS & LEADERSHIP

Executive Board Member | Santa Clara University IEEE

Mar 2024 – Dec 2024

- Collaborated with professors and industry professionals to provide 200+ students with career & academic insights
- Designed quarterly university-wide software & firmware workshops to spark interest in electrical engineering
- Boosted club participation through a weekly newsletter and custom web pages reaching 400+ subscribers

Mentor | Santa Clara University

Aug 2023 – Jul 2024

- Provided structured technical guidance to incoming students, focusing on software engineering best practices, project design, and debugging strategies.
- Led one-on-one mentoring sessions focused on academic planning and career growth

Fist Bump | 3rd Place Hack for Humanity 2024

Feb 2024

- Developed a web platform to streamline student event tracking with a dynamic calendar view built in Dart
- Architected a RESTful backend using Dart's Shelf framework with 5 API endpoints connected to a local datastore for real-time event synchronization

The Marina Project | 3rd Place ACM Winter Challenge

Dec 2023

- Engineered a full-stack mobile application using Flutter and Dart
- Integrated 12+ REST APIs to curate music based on geographic location
- Optimized app responsiveness with data caching for a smooth user experience, improved response time by 30%
- Implemented asynchronous state management and skeleton loading to streamline page transitions and minimize perceptible latency